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Saveetha Institute of Medical And Technical Sciences
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Hospital Patient Record Management System

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA0502 - Database Management System

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY
IN

BTECH AI&DS

Submitted by

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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Hospital Patient Record Management System” has been carried out by Christilla Sweety J S (192421141) and Sandhya K (192424061) under the supervision of Dr. G. Ayyappan and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech. Artificial Intelligence and Data Science (AI&DS) program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Submitted for the Project work Viva-Voce held on _____

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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DECLARATION

We, **K.Sandhya** of the **BTECH AI&DS**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **Hospital Patient Record Management System”** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: chennai

Date: 5-09-2026

Signature of the Students with Names

K.Sandhya

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Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Student 1	Requirements & database	ER diagram, database, core modules	Database & module testing	Introduction, design, ER diagram	[_50%]
Student 2	Coding & interface	UI and system modules	System testing & analysis	Algorithm, testing, conclusion	[_50_%]
cTotal					100%

ABSTRACT

The Hospital Patient Record Management System Using Relational DBMS addresses the engineering problem of efficiently managing large volumes of interconnected hospital data, including patient details, doctor information, appointments, medical records, billing, and payment information. Traditional manual and file-based approaches may result in data duplication, inconsistent records, billing errors, difficulty in retrieving information, and increased administrative workload. Therefore, a reliable and centralized database system is required to improve data accuracy, integrity, accessibility, and overall hospital workflow.

The proposed system uses a Relational Database Management System (RDBMS) to organize hospital information into related tables using primary keys, foreign keys, constraints, and normalization techniques. The Appointment & Billing Module specifically manages patient appointments, doctor schedules, appointment status, bill generation, service charges, payments, and pending amounts. The methodology involves identifying system requirements, designing the relational schema, creating tables, establishing relationships, inserting sample records, applying integrity constraints, and executing SQL queries for data insertion, retrieval, updating, deletion, joining, and billing calculations.

The system is designed to ensure that appointment and billing information remains consistent and can be retrieved quickly when required. The major results include improved organization of patient and appointment records, accurate calculation and tracking of billing information, reduced data redundancy, faster record retrieval, and better management of pending payments. The use of relational database concepts also ensures that related information can be efficiently connected and maintained without unnecessary duplication. The proposed system demonstrates how database engineering principles can be applied to solve practical hospital information-management problems. It provides a structured, reliable, and scalable foundation for managing hospital records and can be further enhanced with authentication, automated notifications, online appointment booking, payment integration, and reporting features. **Keywords :** Hospital Management, Patient Records, Relational DBMS, Appointment Management, Billing Management, SQL

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The Hospital Patient Record Management System Using Relational DBMS is developed to efficiently store and manage hospital information such as patient, doctor, appointment, medical, and billing records. The system uses an RDBMS with tables, primary keys, foreign keys, relationships, and SQL queries to maintain data accuracy and consistency. The Appointment & Billing Module manages patient appointments, doctor schedules, bill generation, payment details, and pending payments. This system reduces manual work, avoids data duplication, provides quick access to records, and improves the overall efficiency of hospital data management.

1.2 Need for the Project

The project is needed to provide an efficient and centralized system for managing hospital patient, appointment, and billing records. Manual and file-based methods can cause data duplication, errors, delays, and difficulty in retrieving information. Patients, doctors, nurses, receptionists, and billing staff are affected by these problems. Existing systems may have poor data organization, limited integration, and difficulty in maintaining accurate records. A relational DBMS provides an engineering solution by using structured tables, relationships, constraints, and SQL queries to improve data consistency, security, and accessibility. This project is therefore significant for reducing manual effort, improving hospital workflow, and enabling faster and more reliable management of appointments and billing information.

1.3 Problem Statement

Hospitals need to efficiently manage large and interconnected patient, doctor, appointment, and billing records while maintaining accuracy and consistency. Manual or separate file-based systems can cause duplicate data, incorrect appointment details, billing errors, delayed retrieval, and inconsistent records. The system must handle multiple interacting factors such as patient

availability, doctor schedules, appointment status, hospital charges, payments, and pending bills.

The proposed solution should use a relational DBMS with primary keys, foreign keys, constraints, normalization, and SQL queries while considering realistic constraints such as data security, integrity, efficient storage, quick retrieval, and avoidance of duplicate records.

The engineering challenge is to design a reliable and scalable database system that provides accurate appointment and billing management with minimum manual effort.

1.4 Project Aim

The main aim of this project is to design and develop an efficient Hospital Patient Record Management System using a Relational DBMS to securely store, organize, and manage patient, doctor, appointment, medical, and billing information.

The system aims to reduce manual work, minimize data duplication and errors, provide quick access to records, and maintain data accuracy and consistency. It also focuses on efficiently managing patient appointments and billing details through well-structured database tables, relationships, constraints, and SQL queries.

1.5 Project Objectives

- To design a relational database structure for managing patient, doctor, appointment, medical, and billing records.
- To develop a centralized database system that stores and retrieves hospital records efficiently.
- To model relationships between patients, doctors, appointments, and billing information using primary and foreign keys.
- To implement SQL operations for adding, updating, deleting, searching, and generating reports from hospital records.
- To test the database using sample records and queries to verify data accuracy, integrity, and consistency.

1.7 Expected Engineering Outcome

The expected engineering outcome is a functional Hospital Patient Record Management System using a Relational DBMS that efficiently stores, manages, and retrieves patient, doctor, appointment, and billing information. The system will provide a structured database with tables, primary keys, foreign keys, relationships, and constraints. It will support appointment scheduling, billing, payment tracking, and record retrieval using SQL queries. The final system is expected to reduce data redundancy, improve data accuracy and consistency, minimize manual work, and provide faster access to hospital information.

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CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, existing hospital management systems, and Relational DBMS technologies were identified through academic journals, textbooks, research papers, and reliable online technical resources. Information related to hospital patient records, appointment management, billing systems, database design, normalization, and SQL was reviewed and compared. Existing database-based hospital systems were studied to understand their features, limitations, and implementation approaches. The review helped identify suitable DBMS concepts, database structures, relationships, and SQL techniques for developing the proposed system.

2.2 Review of Existing Work

Existing hospital management systems and research show that Relational DBMS can effectively manage patient, doctor, appointment, and billing information. Commercial systems provide features such as patient registration, appointment scheduling, medical records, billing, and reporting, but may be costly and complex. Technologies such as MySQL, PostgreSQL, and Oracle support secure data storage, SQL queries, relationships, and data integrity. Existing solutions may have limitations in cost, customization, and complexity. Therefore, this project aims to provide a

simple, efficient, and customizable database system for managing hospital patient records, appointments, and billing.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Record System	Simple and low cost	Time-consuming and prone to errors	Proposed system reduces manual work
File-Based System	Easy to implement	Data duplication and inconsistency	Proposed RDBMS provides organized and consistent data

2.4 Research / Engineering Gap

The proposed Relational Database Management System (RDBMS) provides a centralized and systematic approach to storing, managing, and retrieving hospital information. It organizes patient, doctor, appointment, medical, and billing details into related tables, reducing unnecessary data duplication and improving data consistency. The use of primary keys and foreign keys establishes proper relationships between different records, while constraints help maintain data accuracy and integrity. SQL queries can be used to efficiently insert, update, delete, search, and retrieve information whenever required.

2.5 Proposed Contribution

The proposed project improves existing hospital record management approaches by providing a simple, centralized, and cost-effective relational database system for managing patient, doctor, appointment, medical, and billing information. Unlike manual and file-based systems, it reduces data duplication, improves data accuracy, and provides faster information retrieval through SQL queries. The system also establishes clear relationships between patients, doctors, appointments, and billing records using primary keys, foreign keys, and database constraints.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY 3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Hospital Staff	Manage patient appointments and billing records efficiently.

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Appointment and billing management	Functional	Schedule appointments and generate accurate bills for patients.

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010	Software should provide reliability, security, usability, and maintainability.	The database is designed for accurate, secure, and reliable hospital record management.

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Data Retrieval Time	Less than 3	seconds	High

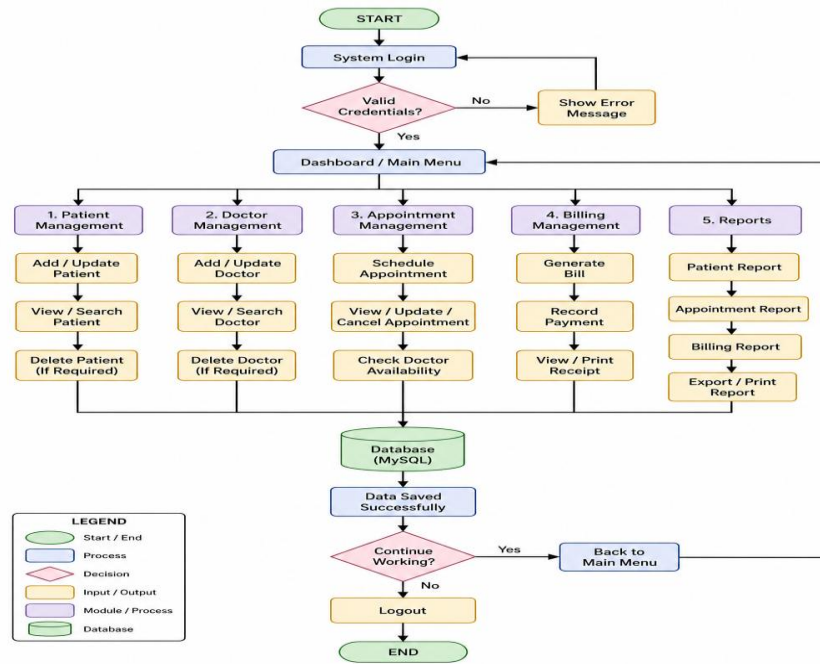
3.4 Alternative Solutions & Evaluation

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	2/5	3/5	5/5
Cost	20%	5/5	4/5	4/5

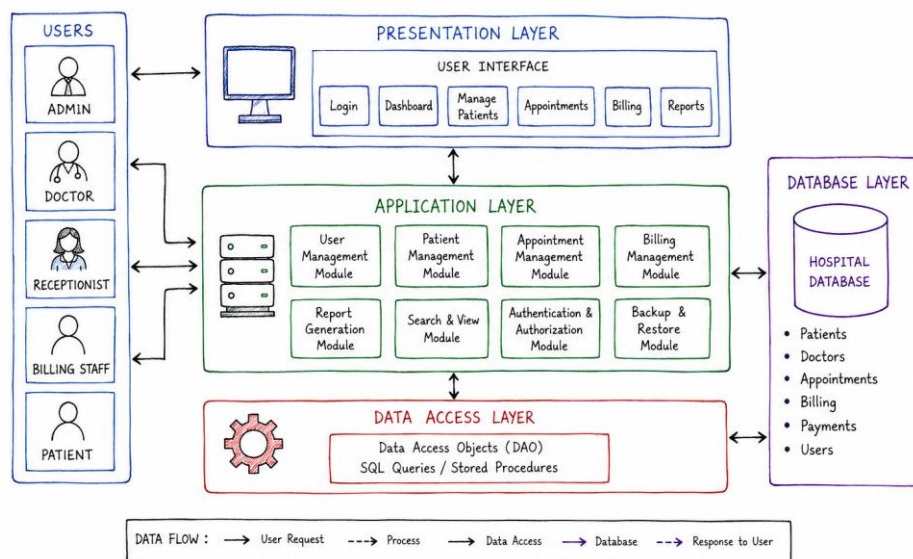
Safety	15%	2/5	3/5	5/5
Sustainability	15%	2/5	3/5	5/5
Reliability	25%	2/5	3/5	5/5

3.5 Selected Design & Detailed Design

Flow Chart :



SYSTEM ARCHITECTURE DIAGRAM



3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Web application	Desktop application	Easy access	Needs browser	Web application selected for accessibility

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce paper records	Digital storage	Digital system selected
Ethics	Protect user information	Access control	Privacy maintained
Health & Safety	Avoid incorrect records	Data validation	Validation included
Security	Prevent unauthorized access	Authentication	Secure access implemented

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect billing data	Medium	High	High	Input validation and testing	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The Appointment Billing Module is developed using an iterative development approach, where the system is divided into smaller modules such as appointment management, patient/customer records, billing, database management, and reporting. Each module is implemented and tested separately before being integrated into the complete system. The development process includes requirement analysis, database design, interface development, implementation, testing, and debugging. The system is designed to provide accurate appointment and billing management while maintaining data security and usability.

Resources Used: Computer/Laptop, Visual Studio Code, Python, Flask, HTML, CSS, MySQL/SQLite database, web browser, sample appointment and billing dataset.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The project is implemented as a software-based web application. Python Flask is used for backend processing, while HTML and CSS are used to create the user interface. The database is used to store appointment, patient/customer, billing, and payment information. Testing is performed using sample records to verify that appointment creation, updating, deletion, searching, and billing calculations work correctly.

[Implementation details — hardware/software/fabrication as applicable.]

Tool / Software / Equipment	Purpose	Stage Used
MySQL / SQLite	Store appointment and billing records	Database Development

MODULE 3

Tables: Appointment & Billing (1:1 Relation)

Appointment Scheduling & Financial Billing Module

Manage slot bookings, doctor consultations, dynamic fee computation (Consultation + Lab + Pharmacy - Discount), and instant PDF invoice generation.

Appointments (9)

Billing & Invoices (7)

+ Book Appointment

Q Search appointment ID, doctor, patient, reason...

All Statuses

Appointment Schedule Ledger (9 entries)

Foreign Keys: Patient_ID & Doctor_ID

Appt_ID	Patient	Consultant Doctor	Date & Time	Reason for Visit	Status	Invoicing Action
A401	P101 FK: P101	D201	2026-01-20 10:00 AM	Cardiac checkup and ECG evaluation	Completed	View Receipt (\$300.00)
A402	P102 FK: P102	D202	2026-02-05 11:30 AM	Severe recurrent migraines evaluation	Completed	View Receipt (\$245.50)
A403	P103 FK: P103	D203	2026-02-20 02:15 PM	Persistent flu and fever symptoms	Completed	View Receipt (\$143.00)

4.3 Test Plan & Verification

Define the test plan *BEFORE* presenting results.

Requirement		Test Method		Acceptance Criterion	
Add Appointment		Enter valid appointment details		Appointment added successfully	
Specification	Required Value	Achieved Value		Status (Pass/Fail)	
Appointment Registration	100%	100%		Pass	

4.4 Results

MODULE 3

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The Appointment Scheduling & Financial Billing Module manages patient appointments, doctor consultations, visit reasons, appointment status, and invoicing in one system. It provides options to book appointments, search records, view billing invoices, and track completed visits, making appointment and billing management faster and more organized.

4.5 Data Analysis & Engineering Judgment

The system was tested using sample appointment and billing records. The results showed that appointments were stored, searched, updated, and billed correctly, with accurate invoice calculations. Minor errors caused by invalid or incomplete input were handled through validation. Overall, the system improved data accuracy, reduced manual errors, and provided faster appointment and billing management.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Manage appointments	Appointment module	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Easy appointment and billing management.
- Reduces manual errors.
- Provides quick record searching.
- Improves data organization and accuracy.

Limitations:

- Depends on internet/system availability.
- Uses sample data for testing.
- Limited to basic billing functions.
- Does not include advanced analytics or online payment integration.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
Student 1	Frontend Development	Designed web pages
Item	Estimated Cost	Actual Cost

Database	₹0	₹0
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CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Appointment Scheduling & Financial Billing Module was developed to simplify appointment management and billing operations. The system successfully supports appointment booking, record management, billing, and invoice generation. Testing and validation showed that the major functions work correctly and reduce manual errors. Overall, the project provides an organized, accurate, and efficient solution for appointment and financial billing management.

5.2 Future Work

In the future, the Appointment Scheduling & Financial Billing Module can be further improved by adding advanced features to make the system more efficient, secure, and user-friendly. Online payment facilities can be integrated to allow patients to pay bills digitally and receive electronic receipts. SMS and email notifications can be added to remind patients about upcoming appointments and pending payments. A mobile application can also be developed to provide convenient access to appointments and billing information from anywhere.

The system can be enhanced with advanced reports and dashboards to analyse appointment trends, revenue, and billing performance. AI-based features can be introduced to predict appointment demand, identify billing patterns, and support better decision-making. Security can also be strengthened through improved authentication, data encryption, and regular database backups. These future improvements will help the system provide a more complete and scalable solution for appointment scheduling and financial billing management.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Learned to develop an appointment and billing management system.
- Gained knowledge of database design and record management.
- Learned to implement billing and invoice calculations.
- Understood input validation and error handling.
- Learned software testing and debugging techniques.

Engineering & Professional Skills Developed:

- Problem-solving and logical thinking.
- Programming and database management skills.
- System design and testing skills.
- Teamwork and task management.
- Technical documentation and presentation skills.

Individual Reflection

Student 1:

I learned how to design and develop the user interface of the system. The most difficult problem was creating simple and clear appointment forms, which was solved through testing and modification. I personally decided to use a simple web-based interface because it was easy to use and maintain. If repeated, I would improve the interface with more responsive design features.

Student 2:

I gained practical knowledge of backend programming and system functionality. The main difficulty was connecting different modules correctly, which I solved by debugging each function separately. I decided to use automated billing calculations to reduce manual errors. If repeated, I would improve the backend structure and add more validation.

Student 3:

I learned how to create and manage the database for appointment and billing records. The most difficult problem was maintaining accurate and organized data, which was solved using proper

database tables and validation. I selected a database-based approach because it provides reliable data storage. If repeated, I would improve database security and backup facilities.

Student 4:

I learned the importance of software testing and documentation during the project. The main challenge was identifying and correcting errors in different modules, which was solved through repeated testing. I personally focused on verifying the system functions before final submission. If repeated, I would include more test cases and automated testing.

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APPENDICES

Appendix A – Detailed Calculations

This appendix presents the detailed calculations used in the Appointment Scheduling & Financial Billing Module. It includes sample calculations for consultation charges, service charges, discounts, taxes, and final invoice amounts. The calculations demonstrate how the system automatically determines the total billing amount based on the services selected and the applicable charges. These examples provide supporting evidence that the billing process is consistent and reduces manual calculation errors.

Appendix B – Engineering Drawings / Schematics

This appendix contains the engineering and system design diagrams prepared during the development of the project. It includes the system architecture diagram, database/ER diagram, flowchart, use-case diagram, and module structure. These drawings show how users interact with the system, how appointment and billing information flows through different modules, and how data is stored and processed. The diagrams support the design decisions made during implementation.

Appendix C – Source Code / Code Repository Information

This appendix provides information about the source code developed for the project. Important code excerpts related to appointment registration, record searching, billing calculation, database

connectivity, validation, and invoice generation may be included in the report. Since including the complete source code would make the report unnecessarily lengthy, the full source code is maintained separately in the approved project repository. The repository contains the required application files, database files, templates, stylesheets, and supporting resources.

Appendix D – Datasheets

This appendix is **not applicable** to the project because the Appointment Scheduling & Financial Billing Module is a software-based system and does not use physical electronic or hardware components. Therefore, hardware datasheets, component specifications, and fabrication documents are not required. The software tools and technologies used in the project are documented in the main report and references section.

Appendix E – Experimental / Raw Data

This appendix contains the sample and raw data used during system development and testing. It includes sample patient/customer details, appointment records, service information, billing entries, payment details, and test inputs and outputs. The data was used to verify whether the system could correctly store, retrieve, update, and process appointment and billing information. Sample data was used instead of real personal or patient information to maintain privacy.

Appendix F – Ethics / Institutional Approval

This appendix documents the ethical considerations related to the project. Since the system was developed using sample or dummy data and does not involve real patients, human experiments, or collection of sensitive personal information, formal institutional approval was not required for the development and testing process. The project nevertheless considers important principles such as data privacy, secure access, accurate billing, responsible data handling, and prevention of unauthorized access.

Appendix G – Project Meeting / Progress Records

This appendix contains records of the project development activities and team progress. It includes information about team meetings, task allocation, discussions, development milestones, testing activities, problems identified, and solutions implemented. These records demonstrate that the project was developed systematically and that responsibilities were distributed among team members. Progress records also provide evidence of continuous monitoring and completion of project activities.

Appendix H – User/Industry Feedback

This appendix contains feedback obtained from users, classmates, faculty members, or evaluators during the development and testing of the system, where applicable. The feedback mainly focuses on system usability, appointment management, billing accuracy, interface design, and ease of navigation. Suggestions received during evaluation were considered to improve the clarity, functionality, and user experience of the system. If formal industry feedback was not obtained, this appendix may be marked as not applicable.

Appendix I – Publications / Patents / Competitions / Product Outcomes

This appendix records any publications, patents, competitions, demonstrations, or product outcomes associated with the project, where applicable. For the current project, no patent or formal publication has been produced. If the project is later presented in a competition, seminar, exhibition, or academic publication, the relevant certificates, papers, screenshots, or supporting evidence can be added to this appendix.

Appendix J – Individual Contribution Evidence

This appendix provides evidence of the individual contributions made by each team member. It

includes assigned responsibilities, completed modules, programming activities, database development, interface design, testing, documentation, presentation work, and repository contributions. The contribution records help demonstrate that each member participated actively in the project and completed their assigned tasks. Screenshots of code commits, module development, testing results, meeting records, and other relevant evidence can be included to support the contribution claims.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Ch. 4 (4.8)